Sre Cibi Varsshan K

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EDUCATION

Vellore Institute of Technology, Chennai | Expected 2027
B.Tech Computer Science and Engineering (Artificial Intelligence and Machine Learning)
Relevant coursework: Data Structures and Algorithms, Machine Learning, Computer Vision.

SUMMARY

Android developer building offline-first apps: a Kotlin and Jetpack Compose LLM assistant with on-device inference through JNI and a Room/SQLite memory layer, contributions to a Flutter team app with a Bluetooth LE relay, and Linux tooling for Android OS installation.

PROJECTS

XrivenixAI - offline LLM assistant for Android | Kotlin, Jetpack Compose, Room, JNI | Sep 2026
github.com/xerfzy/XrivenixAI

- Built an Android app (minSdk 24, arm64-v8a) that runs GGUF language models on the device CPU through llama.cpp, with no account, backend or cloud inference; the network is used only for explicit model downloads.
- Wrote the C++ JNI bridge (NDK and CMake) for streaming generation, grammar-constrained extraction, embeddings and KV-cache reuse, and serialised all native calls on a single-thread coroutine dispatcher so only one model is ever loaded.
- Designed the persistence layer in Room with schema migrations 1 to 4 and FTS5 full-text tables; retrieval merges embedding similarity, full-text and entity matches, and validation code de-duplicates and versions extracted facts by topic.
- Implemented resumable model downloads with WorkManager and a foreground notification (HTTP Range requests, SHA-256 check, atomic rename) and device-aware model suitability from RAM, storage and CPU features.
- Built Compose (Material 3) screens with ViewModel and StateFlow: streaming chat with stop, Markdown and LaTeX rendering, model manager, memory browser with search and editing, settings.
- Wrote 30 JVM unit tests (JUnit 4) and 29 instrumented tests and benchmarks; measured about 5-6 tokens/s generation and about 1.8 s to first token with KV reuse for a 3B model on a Snapdragon 782G phone.

AgriEdge - offline farming assistant (3-person team) | Flutter, Dart, Bluetooth LE, Node.js | Oct 2026
github.com/xerfzy/AgriEdge

- Fixed the Android build and made the team's Bluetooth LE alert mesh work end to end on Android phones: wrote the BLE framing layer, handled advertising-set limits and acknowledged packets only after storing them, with P2P regression tests.
- Added live Agmarknet market prices through the data.gov.in API, photo sharing over BLE, a farm-supplies buy and sell flow, and a UI redesign.
- Built the password-protected Express control panel and deployed the server to Render so connected phones relay official advisories to offline ones.

eos-oneplus-nord - Android OS installation tooling | Bash, Linux, adb, fastboot | Oct 2026
github.com/xerfzy/eos-oneplus-nord

- Wrote scripts that check device identity, Android version, build signing, battery, OEM-unlock and account state over adb, verify the /e/OS image and installer by SHA-256, and gate flashing on fastboot product and unlock state.
- Used the tooling to install /e/OS (Android 16) on a OnePlus Nord and documented Magisk rooting.

RESEARCH

Indian Patent Application IN202641060337 A1 (published application, not granted) | May 2026
"A System for Swarm of Drone for Crowed Behaviour Analysis" (title as published). Application No. 202641060337, filed 12 May 2026, published 22 May 2026. Inventors: Sre Cibi Varsshan K, D Kavitha. Applicant: Vellore Institute of Technology, Chennai.

TECHNICAL SKILLS

Android: Kotlin, Jetpack Compose (Material 3), ViewModel, StateFlow, Coroutines, Room (migrations, FTS5), WorkManager, foreground services, Gradle (Kotlin DSL)
Native: C++, JNI, Android NDK, CMake, llama.cpp
Cross-platform: Flutter, Dart, Riverpod, Drift/SQLite, Bluetooth LE
Testing and tools: JUnit 4, instrumented tests, Git, adb, fastboot, Linux
Other languages: Python, C#, Bash